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MEDICAL DEVICE INNOVATION WORKSHOP

SCAN TO REGISTER

18th March 2026

#19 Washington Square North
New York, NY 10011

جامعة نيويورك أبوظبي
NYU ABU DHABI

NYU | TANDON SCHOOL OF ENGINEERING

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Sponsored by the 19 Washington Square North Fellowship 2025-2026

MEDICAL DEVICE INNOVATION WORKSHOP

OVERVIEW

We invite you to attend our interdisciplinary workshop that brings together engineers, clinicians, and researchers from NYU Tandon, NYU Abu Dhabi, NYU Langone Health, University of Maine, and University of Michigan to explore the future of medical devices and implants

The workshop will focus on innovative approaches to glaucoma treatment and beyond, highlighting cutting-edge research in microfluidics, computational modeling, and translational device design.

Through expert talks and discussions, participants will gain insight into how engineering and medicine can collaborate to improve patient outcomes.

The discussion will also cover medical implants across ophthalmic, gastrointestinal, and neurological applications, fostering new connections and collaborations.

MEDICAL DEVICE INNOVATION WORKSHOP

MESSAGE FROM THE CHAIRS

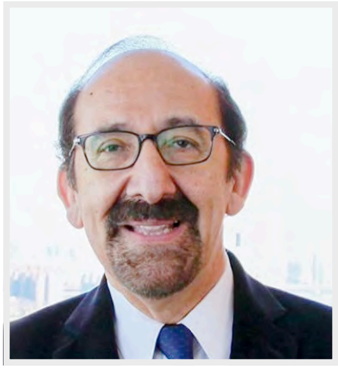
Dear Colleagues,

We welcome you to this workshop supported by the 19 Washington Square North Fellowship. It is dedicated to the latest advancements in medical device technology.

Our focus this year bridges three critical frontiers of healthcare: glaucoma, gastrointestinal disorders, and neuronal diseases. While these fields may seem disparate at first glance, they are united by the urgent need for high-precision diagnostics and minimally invasive interventions.

Our goal for this workshop is simple but ambitious: to spark the cross-disciplinary collaborations that will define the next decade of medical innovation. Whether you are here to share groundbreaking data or to find your next technical partner, we encourage you to engage deeply, challenge the status quo, and help us push the boundaries of what is possible in patient care and solve unmet clinical needs.

Rafael (Yong-Ak) Song
NYU Abu Dhabi



Iskender Sahin
NYU Tandon

MEDICAL DEVICE INNOVATION WORKSHOP

PROGRAM AGENDA

Wednesday, 18th March 2026

TIME	ACTIVITY / PRESENTATION
09:00 - 09:15 AM	Welcome Remark by Iskender Sahin and Rafael Song
09:15 - 10:00 AM	Andreas Hielscher, NYU Tandon *Seeing the Invisible: Real-Time Vascular Imaging with Wearable Optical Tomography*
10:00 - 10:30 AM	Sohmyung Ha, NYU Abu Dhabi *Intra-body power transfer for wearable devices*

10:30 - 11:00 AM : COFFEE BREAK

TIME	ACTIVITY / PRESENTATION
11:00 - 11:30 AM	Giovanna Guidoboni, University of Maine *From the Blackboard to the Clinic: digital twinning and artificial intelligence for personalized medicine*
11:30 - 12:00 PM	Manjool Shah, The University of Michigan *The GDD Revolution: From Refractory Cases to Primary Care*
12:00 - 12:30 PM	Nurbergen Aitmukhanbetov, NYU Abu Dhabi *Title: Development of a New Minimally Invasive Glaucoma Implant*

12:30 - 02:00 PM : LUNCH

Continued...

MEDICAL DEVICE INNOVATION WORKSHOP

PROGRAM AGENDA

Wednesday, 18th March 2026

TIME	ACTIVITY / PRESENTATION
02:00 - 02:30 PM	Shy Shoham, NYU School of Medicine and Tech4Health *Engineering at the interface: from Neurotech development to Tech4Health*
02:30 - 03:00 PM	Rafael Song, NYU Abu Dhabi *Microphysiological Systems for Growing Organs and Organoids*
03:00 - 03:30 PM	Khalil Ramadi, NYU Abu Dhabi *Devices you can eat: Speaking with the body through the gastrointestinal tract*
03:30 - 04:00 PM	Sefy Paulose Joshi, NYU Langone *Under Pressure: The Evolution of Glaucoma Devices*
04:00 - 04.30 PM	Iskender Sahin, NYU Tandon *Computational Study of Gene Therapy into the Retina*

04:30 - 05:00 PM : BREAK

TIME	ACTIVITY / PRESENTATION
05:00 - 06:00 PM	**Bridging the Bedside & the Bench: A MedTech Panel Discussion** *Panelists:* › Manjool Shah (University of Michigan) › Shy Shoham (NYU Langone) › Giovanna Guidobonni (University of Maine) › Andreas Hielscher (NYU Tandon) › Sefy Paulose Joshi (NYU Langone) *Moderator:* › Rafael Song (NYU Abu Dhabi)

06:00 - 08:00 PM : DINNER

PRESENTATIONS

Abstracts & Bios

MEDICAL DEVICE INNOVATION WORKSHOP

SEEING THE INVISIBLE: REAL-TIME VASCULAR IMAGING WITH WEARABLE OPTICAL TOMOGRAPHY

Andreas Hielscher

Professor and Chair of Biomedical Engineering, NYU Tandon School of Engineering

BIO

Professor Andreas H. Hielscher received his PhD in Electrical and Computer Engineering from Rice University in Houston, Texas. After a postdoctoral fellowship at Los Alamos National Laboratory in New Mexico, he joined the faculty at SUNY Downstate Medical Center. In 2001, he accepted an offer from Columbia University in New York City, where he became Director of the Biophotonics and Optical Radiology Laboratory. At Columbia, he held professorships in the Departments of Biomedical Engineering, Electrical Engineering, and Radiology.

In the summer of 2020, he moved across town to lead the newly established Department of Biomedical Engineering at New York University (NYU). His research centers on the development of advanced optical tomography systems. He applies this imaging technology to clinical challenges in arthritis, vascular diseases, and breast cancer, and is currently pioneering an optical brain-machine interface. Professor Hielscher has published more than 300 scientific papers, authored several book chapters, and holds over 20 patents.

His research has been continuously supported by institutes within the National Institutes of Health (NIAMS, NIBIB, & NCI), DARPA, and other national and international funding bodies. Since 2013, he has been a Fellow of the American Institute for Medical and Biological Engineering (AIMBE).

ABSTRACT

Vascular Optical Tomographic Imaging (VOTI) is an emerging non-invasive imaging modality that enables real-time, three-dimensional mapping of key physiological parameters such as oxy-hemoglobin (HbO_2), deoxy-hemoglobin (Hb), total hemoglobin (THb), oxygen saturation (StO_2), and water content in deep tissue. By illuminating the body with near-infrared light and capturing transmitted and reflected signals, model-based iterative algorithms—now increasingly powered by artificial intelligence—reconstruct volumetric functional maps that reveal vascular health and disease.

Over the past decade, my lab has translated this technology from benchtop prototypes to clinically viable systems that target a wide range of applications, including brain function monitoring, breast and prostate cancer diagnosis, peripheral artery disease assessment in diabetic patients, and the evaluation of autoimmune and joint disorders such as lupus arthritis.

In this presentation, I will share recent clinical results and explore how VOTI is becoming a practical bedside and even at-home tool, thanks to advancements in flexible electronics and textile-integrated optoelectronics. These innovations enable wearable systems that promise to democratize vascular imaging by enhancing patient comfort, expanding monitoring beyond the clinic, and supporting longitudinal data collection for personalized care.

INTRA-BODY POWER TRANSFER FOR WEARABLE DEVICES

Sohmyung Ha

Associate Professor of Electrical and Bioengineering, NYU Abu Dhabi, Abu Dhabi

BIO

Professor Sohmyung Ha received the B.S (summa cum laude) and the M.S. degrees in Electrical Engineering from the Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Korea, in 2004 and 2006, respectively. From 2006 to 2010, he worked at Samsung Electronics as an integrated circuit designer for commercial multimedia devices. After this industry career, he returned to academia as a Fulbright Scholar and obtained the M.S. and Ph.D. degrees in Bioengineering with the Engelson Best Ph.D. Thesis Award for Biomedical Engineering from the Department of Bioengineering, University of California, San Diego, La Jolla, CA, USA. Since 2016, he has been with New York University Abu Dhabi, Abu Dhabi, UAE, where he is now Associate Professor of Electrical Engineering and Bioengineering.

He currently serves as an associate editor of IEEE Transactions on Biomedical Circuits and Systems and IEEE Open Journal of the Solid-State Circuits Society. He is a member of the Analog Signal Processing Technical Committee (ASP TC) and the Biomedical and Life Science Circuits and Systems Technical Committee (BioCAS TC) of the IEEE Circuits and Systems Society (CASS). He also served as a member of the IMMD Subcommittee of the International Technical Program Committee (ITPC) of the International Solid-State Circuits Conference (ISSCC).

ABSTRACT

Intra-body power transfer (IBPT) is a technique that delivers power to distributed wearable devices through the human body.

It has emerged as an alternative or even a complementary method to traditional energy harvesting (EH) and inductive wireless power transfer (WPT) as it can supply power to wearable devices regardless of their locations on the body. Moreover, IBPT enables the miniaturization of wearable sensor nodes by physically separating the bulky power source from the sensor load.

It allows the power source to be placed at an optimal location on the body for EH or WPT. In this talk, basic principles, system modeling, challenges, and techniques of IBPT will be presented.

MEDICAL DEVICE INNOVATION WORKSHOP

FROM THE BLACKBOARD TO THE CLINIC: DIGITAL TWINNING AND ARTIFICIAL INTELLIGENCE FOR PERSONALIZED MEDICINE

Giovanna Guidoboni

Professor of Electrical and Computer Engineering and Dean of Engineering and Computing, University of Maine

BIO

Giovanna Guidoboni is the Inaugural Dean of Maine College of Engineering and Computing at the University of Maine, and Interim Vice President for Research at the University of Maine and the University of Maine at Machias. Her research focuses on mathematical modeling and data science in engineering and life sciences, with collaborations in ocular blood flow, urinary tract function, and noninvasive health monitoring. Her work has been supported by the National Science Foundation, National Institutes of Health, and the European Commission. Giovanna co-founded Artificial Intelligence in Vision and Ophthalmology and serves as its co-Editor-in-Chief and Managing Editor. Additionally, she is the founder of Gspace LLC, a consulting firm specializing in modeling and computational solutions.

Dr. Guidoboni holds a PhD in Mathematics and MS and BS in Engineering of Materials from the University of Ferrara (Italy). Before joining the University of Maine in 2023, she held faculty positions at multiple institutions in the United States and in France. She is Professor of Electrical and Computer Engineering, and faculty in the Graduate School of Biomedical Science and Engineering at the University of Maine, and Adjunct Professor of Ophthalmology at Icahn School of Medicine at Mount Sinai (New York, NY). She is an elected member of the European Academy of Sciences and Arts.

ABSTRACT

Digital twinning and artificial intelligence combine mechanism-driven and data-driven models in a complementary and synergistic way to enable personalized medicine.

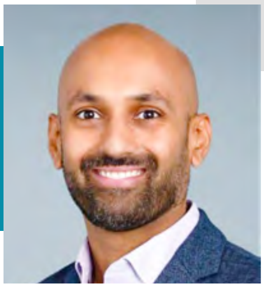
Mechanism-driven models are based on the principles of physics and physiology and allow for identification of cause-to-effect relationships among interplaying factors in a complex system. While invaluable for causality, mechanism-driven models are often based on simplifying assumptions to make them tractable for analysis and simulation; however, this often brings into question their relevance beyond theoretical explorations.

Data-driven models aim at extracting information and knowledge from data. These approaches are naturally interdisciplinary, as it bridges fundamental techniques of data analysis, typically developed by mathematicians, statisticians and computer scientists, with the needs of actionable insights that are specific to the particular application domain.

The combination of mechanism-driven and data-driven models allows us to harness the advantages of both approaches, as mechanism-driven models excel at interpretability but suffer from a lack of scalability, while data-driven models are excellent at scale but suffer in terms of generalizability and insights for hypothesis generation. This combined, integrative approach represents the pillar of the interdisciplinary approach to data science that will be discussed in this talk, with applications spanning from glaucoma research to cardiovascular monitoring and physiology of the lower urinary tract (LUT).

MEDICAL DEVICE INNOVATION WORKSHOP

THE GDD REVOLUTION: FROM REFRACTORY CASES TO PRIMARY CARE



Manjool Shah

Clinical Associate Professor of Ophthalmology and Visual Sciences and Associate Chair of Innovation, University of Michigan

BIO

Manjool Shah serves on faculty at the Kellogg Eye Center at the University of Michigan in the Division of Cataract, Glaucoma and Anterior Segment Disease, where he also serves as Associate Chair of Innovation.

Dr. Shah completed his residency training at the Casey Eye Institute and fellowship training in Glaucoma and Advanced Anterior Segment Surgery at the University of Toronto.

He is active in teaching residents and fellows, as well as through his work on various clinical committees with the American Society of Cataract and Refractive Surgeons (ASCRS), American Academy of Ophthalmology (AAO), and American Glaucoma Society (AGS).

Furthermore, Dr. Shah is passionate about global ophthalmology, and has lectured and taught surgery in Nigeria, Peru, Nepal, Kenya, and Jamaica. His specialized interests include performing and teaching MIGS surgeries as well as traditional glaucoma surgeries, as well as the management of complex anterior segment challenges such as dislocated IOL's, subluxated or complex cataracts, and pupil and iris anomalies.

ABSTRACT

Once the "last resort" for refractory glaucoma, Glaucoma Drainage Devices (GDDs) are now utilized earlier in surgical management due to enhanced design and predictable outcomes. This talk examines the shift from traditional hardware toward next-generation, titratable technology.

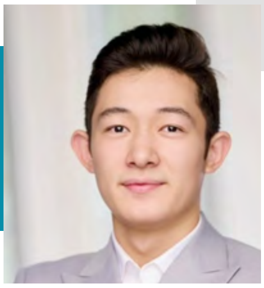
Key Highlights: * Comparison of Standards: Clinical nuances between valved (Ahmed) and non-valved (Baerveldt) devices.

Design Innovations: The rise of low-profile implants like the Ahmed ClearPath and the small-lumen Paul Glaucoma Implant (PGI) to minimize hypotony and improve bleb stability. The Future of Flow: A look at "smart" titratable devices and bio-inert materials designed to reduce the hypertensive phase and long-term fibrosis.

Conclusion: As GDDs become more adjustable and less invasive, they are redefining the boundary between traditional filtering surgery and MIGS, offering a more personalized approach to intraocular pressure control.

MEDICAL DEVICE INNOVATION WORKSHOP

DEVELOPMENT OF A NEW MINIMALLY INVASIVE GLAUCOMA IMPLANT



Nurbergen Aitmukhanbetov

**Research Assistant at CENTMED,
NYU Abu Dhabi, Abu Dhabi**

BIO

Nurbergen Aitmukhanbetov holds a Bachelor of Science in Mechanical Engineering from New York University Abu Dhabi. He specialized in Computational Fluid Dynamics (CFD), Finite Element Modeling (FEM), and fluid-structure interaction (FSI) simulations, and medical device design.

His current research focuses on developing a next-generation glaucoma drainage implant, using computational analysis and experimental verifications to improve fluid control and optimize surgical procedures and postoperative outcomes.

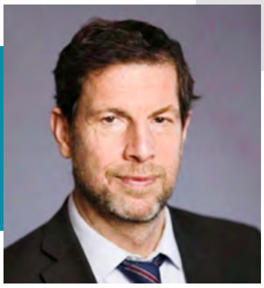
Nurbergen's broader interests also include application of robotics, AI, and Machine Learning toward biomedical technologies.

ABSTRACT

Glaucoma is a leading cause of irreversible blindness worldwide, primarily due to elevated intraocular pressure (IOP). Although conventional glaucoma drainage devices can effectively reduce IOP in severe cases, achieving long-term and reliable IOP control remains a challenge. These traditional devices often come with high levels of surgical invasiveness and significant postoperative complications, such as hypotony. This clinical issue has sparked the recent development of minimally invasive glaucoma surgery (MIGS) devices.

In this presentation, we will introduce a MIGS device featuring several innovative attributes that enhance glaucoma care. This implant aims to effectively lower glaucomatous IOP while maintaining balanced aqueous humor (AH) outflow. Additionally, it is designed to prevent early hypotony and minimize surgical trauma. Our device is minimally invasive, provides stable IOP control with reduced postoperative management requirements, and lays a solid foundation for clinical translation.

ENGINEERING AT THE INTERFACE: FROM NEUROTECH DEVELOPMENT TO TECH4HEALTH



Shy Shoham

Professor of Neuroscience and Ophthalmology, NYU School of Medicine and Tech4Health

BIO

Shy Shoham is a Professor of Neuroscience and of Ophthalmology at NYU Grossman School of Medicine, and director of NYU's Tech4Health institute. His lab develops acoustic, photonic and computational tools for bidirectional neural interfacing, and is exploring translational applications of transcranial ultrasound stimulation.

He holds a Physics BSc (Tel-Aviv University), a PhD in Bioengineering (University of Utah) and was a Lewis-Thomas postdoctoral fellow at Princeton University. Dr. Shoham is a Fellow of SPIE; he serves on the editorial board of Neurophotonics and has co-edited the Handbook of Neurophotonics.

ABSTRACT

In this talk I will describe our innovative and highly collaborative approach to cutting-edge technological R&D at the interface of biomedicne and advanced engineering and physics. Drawing our inspiration from Bell Labs' "idea factory", we have developed a unique model for NYU Tech4Health.

In our approach, PI-led research labs and a dedicated engineering team work side-by-side to address unmet needs and unsolved problems across NYU Langone Health's mission areas. I will illustrate how this approach could potentially circumvent hidden barriers and lead to significant, non-trivial innovations across both healthcare and discovery.

MICROPHYSIOLOGICAL SYSTEMS FOR GROWING ORGANS AND ORGANOIDS

Yong-Ak (Rafael) Song

Program Head of Bioengineering and Professor of Mechanical and Bioengineering

Director of the Research Center for Translational Medical Devices (CENTMED),
NYU Abu Dhabi, Abu Dhabi

BIO

Yong-Ak (Rafael) Song received an MEng degree in Mechanical Engineering from the RWTH Aachen University of Technology in 1993 and a Ph.D. degree from the School of Mechanical Engineering, RWTH Aachen University in 1996.

He was a senior research scientist at the Korea Institute of Science and Technology (KIST) until 2001. He moved to Boston and worked at MIT until 2012. He is currently a professor of mechanical and bioengineering in the Division of Engineering at New York University in Abu Dhabi (NYUAD).

He holds joint appointments in the Department of Chemical and Biomolecular Engineering and the Department of Biomedical Engineering at New York University. He is also the program head of the Bioengineering Program at NYUAD and Director of the Research Center for Translational Medical Devices (CENTMED). His main research interests are in various aspects of micro- and nanoscale bioengineering, including biosensors, point-of-care diagnostics, organ-on-a-chip and organoid systems, and biomimetics.

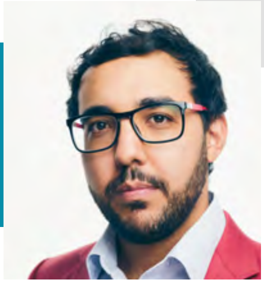
ABSTRACT

Organ-on-a-chip (OOC) technologies have emerged as engineered microfluidic platforms that aid in drug development, disease modeling, and personalized medicine. These systems replicate the functions of various organs at a microscale, thereby eliminating the need for animal testing.

By using living cells in controlled environments, OOC platforms mimic key aspects of organ functions. Building upon this concept, organoid-on-a-chip (OrgOC) platforms represent an advanced version of OOC technologies. They integrate three-dimensional constructs derived from induced pluripotent stem cells (iPSCs) directly within engineered microfluidic chip architectures.

In this talk, I will showcase some of our efforts at the Micro- and Nanoscale Bioengineering Lab at NYUAD to develop an organ- and organoid-on-chip device that replicates the functions of various human organs. By scaling up this microphysiological system, we envision creating a portable organ perfusion system that could support various clinical transplantation applications, particularly for liver and kidney transplants.

DEVICES YOU CAN EAT: SPEAKING WITH THE BODY THROUGH THE GASTROINTESTINAL TRACT



Khalil Ramadi

Assistant Professor of Bioengineering, NYU Abu Dhabi, Abu Dhabi

BIO

Prof. Khalil Ramadi is an Assistant Professor of Bioengineering and Director of the Laboratory for Advanced Neuroengineering and Translational Medicine (LANTRN) at New York University (NYU) Abu Dhabi. His work focuses on developing new tools and technologies for treatment a variety of neurologic, endocrine, and immune disorders.

Prof. Ramadi has been named a TED junior fellow, CIFAR Global Scholar, and MIT Technology Review Innovator Under 35 (MENA), and received multiple honors including the NIH F32 Ruth Kirschstein Postdoctoral Fellowship, BMES Career Development Award, and a NASA Aeronautics Scholarship. He holds a Ph.D. in Biomedical Engineering and Medical Physics from MIT, a M.S. in Mechanical Engineering from MIT, and B.S. degrees in Mechanical Engineering and Bioengineering from the Pennsylvania State University. He is also a board member and former co-Director of MIT Hacking Medicine, a group dedicated to enabling multi-disciplinary health entrepreneurship worldwide.

ABSTRACT

The gastrointestinal tract is a unique point of intersection of digestive, metabolic, nervous, and immune system. Interfacing with the GI tract is quite challenging, given the harsh, motile, and variable environments of the stomach and intestine. This talk will present a variety of engineering approaches for the design and manufacturing of ingestible devices to interface with GI tissue.

We illustrate a variety of clinical applications for such devices, including neuromodulation, assessment of GI motility, and sampling of the gut microbiome. Such ingestible platforms could be a novel approach towards interrogating physiology, diagnosing disease, and delivering therapy non-invasively through the GI tract.

UNDER PRESSURE: THE EVOLUTION OF GLAUCOMA DEVICES

Sefy Paulose Joshi

**Assistant Professor of Ophthalmology,
NYU Langone Health, New York**

BIO

Assistant Professor of Ophthalmology at NYU Langone and a glaucoma-trained ophthalmologist -- she completed her ophthalmology residency at the New York Eye and Ear Infirmary of Mount Sinai and pursued a glaucoma fellowship at Wills Eye Hospital. She is dedicated to advancing glaucoma care through clinical practice, surgical innovation and education.

My practice is dedicated to a wide array of eye conditions, with a specific focus on glaucoma and cataract care. I address all forms of adult glaucoma— including open-angle, closed-angle, normal tension, and secondary types such as pigmentary, pseudoexfoliative, neovascular, uveitic, traumatic, and more.

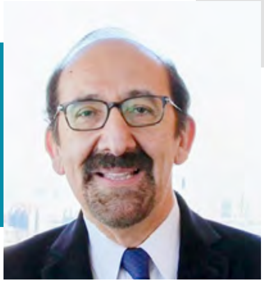
I am skilled in performing routine to advanced cataract surgeries for patients with and without glaucoma and offer a full spectrum of glaucoma treatments, including laser-based, incisional surgeries, and minimally invasive glaucoma surgeries (MIGS).

ABSTRACT

Glaucoma surgery has evolved from traditional filtration procedures to advanced implantable and minimally invasive devices.

From the surgeon’s perspective, this talk reviews past and present approaches, highlights ongoing challenges, and explores the future of next-generation glaucoma valves designed to improve outcomes and patient care.

COMPUTATIONAL STUDY OF GENE THERAPY INTO THE RETINA



Iskender Sahin

Industry Professor of Mechanical and Aerospace Engineering, New York University - Tandon School of Engineering

BIO

Professor Iskender Sahin received his degrees from Istanbul Technical University (BS), The University of Michigan (M.S.) and Virginia Tech (PhD). His academic positions included Virginia Tech, Western Michigan University and the Polytechnic Institute of Brooklyn – now New York University Tandon School of Engineering. He has spent several Summer Faculty Fellowships at NASA Lewis (currently NASA Glenn), Naval Coastal Systems Center and a Naval Research Laboratory Post-Doctoral

Fellowship: His research spans from Stability and Transition in 3D Boundary layers to flow around submarines and ships, including analyzing flow around an extraterrestrial Titan submarine. His recent research activities include the application of CFD for predicting aneurism and improving glaucoma valves and gene therapy for eye diseases.

His research funding has come from the US Navy, NASA and NSF as well as industrial sponsors. Professor Sahin is active in publications, professional societies and is a Fellow of American Society of Mechanical Engineers (ASME)

ABSTRACT

Inherited and acquired retinal disease are the number one cause of blindness in the world. In search of an effective resolution gene therapy represents a new frontier in ophthalmology as opposed to symptomatic management.

The eye's structure presents a convenient target for localized delivery of the gene protein with minimal exposure by either subretinal or intravitreal injection. We present a unique computational model of a gene therapy using transfer of proteins into the retina with a subretinal injection. The approach utilizes the observation as eye being modeled as a native dipole using Computational Fluid Dynamics (CFD) .

Initial computational attempt approximates the "gene" as a particle and utilizes a Lagrangian tracing. However, this approach oversimplifies the physics as the drug (gene) is a concentration of a "bulk fluid" introduced into the Aqueous Humor (AH) in the eye instead of a few finite-size particles.

The target for the drug is the vital inner layer that is a photoreceptor. Several configurations of the injection and its locations are investigated in addition to the dipole positions and strengths that mimics the actual drug delivery mechanism.

Improvements are suggested to the geometry of the retina more realistically in addition to an iterative study of flow parameters. The temperature effects as related to the vitreous and the gene and the wall oscillations as if influences the fluid viscosity will also be considered in future research.

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